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Reproducibility of Honey Taste and Odor Typicality Assessment: A Blind Consumer Panel Versus the Verdicts of Four Laboratories

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Abstract

Organoleptic assessment can decide whether a honey batch is admitted to the market, although the reproducibility of taste assessment is difficult to verify. This study examined interlaboratory and consumer–laboratory agreement on typicality verdicts. Seven honey batches were assessed in a blind consumer panel (138 respondents, 966 ratings) and, in parallel, by four laboratories. Typicality in the blind consumer panel correlated with the number of laboratories without objections ($\rho = 0.76$), but this agreement rested on one sample (P3), the only one the panel classified as typical. After its exclusion, the panel did not discriminate between the remaining six samples ($p = 0.274$), and the typicality of the two samples flagged by most laboratories did not differ significantly from 0.5. A single consumer agreed with a laboratory on whether a sample was not typical for honey to a similar degree as with another consumer ($\kappa = 0.10$ and 0.08). Agreement among the four laboratories was fair on the binary scale ($\kappa = 0.30$) and slight on the three-level scale ($\kappa = 0.13$). A single consumer rating reproduced the laboratory majority verdict with AUC = 0.57. Taste typicality assessment identifies extreme samples, but its reproducibility for borderline samples is limited, and as a stand-alone criterion it does not meet the objectivity required for commercial quality assessment.

Keywords: honey; organoleptic assessment; blind consumer panel; interlaboratory agreement; taste typicality; commercial quality

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1. Introduction

The quality of honey is assessed not only by instrumental analysis but also by the senses of sight, smell and taste. Organoleptic assessment is inexpensive and quick to perform, and its result has practical consequences, because it may decide whether a batch is admitted to the market. At the same time, it is the most difficult part of the assessment to standardize, since it relies on human judgment rather than on an instrument reading.

The legal framework for this assessment is set by European Union law, national regulations and international standards.

The commercial quality of honey placed on the market in the European Union is defined by Directive 2001/110/EC [1]. The directive is transposed into the law of the Member States, and outside the Union the reference point is the Codex Alimentarius standard CXS 12-1981 [2]. Besides instrumentally determined parameters, both documents contain organoleptic requirements. Honey must not have any foreign taste or odor, and a floral (varietal) name may be used only if the product has the organoleptic, physicochemical and microscopic characteristics of the declared source.

Polish regulations impose a broader requirement. The regulation on the commercial quality of honey [3] considers honey compliant if it "has an odor and taste typical of the given honey variety". A finding of atypical taste, however, requires a comparison with a reference for the variety, which none of these acts defines. Descriptive sheets of the main European unifloral honeys are available in the literature [4], and a methodology of organoleptic assessment has been published within the International Honey Commission (IHC) [5]. These descriptions do not, however, have the status of a legal reference. At the same time, Article 5 of the directive [1] prohibits Member States from adopting national provisions on honey not provided for by the directive.

In Poland, the method of organoleptic assessment is laid down in the regulation on analytical methods for honey [6]. The regulation only states that color, consistency, taste and odor are examined with the senses of sight, smell and taste, and that the results are compiled in a table. It specifies neither the number nor the qualifications of the assessors, nor a reference material, nor a criterion for classifying a taste as atypical. Directive (EU) 2024/1438, applicable from 14 June 2026 [7], provides for the adoption of harmonized methods for detecting honey adulteration, but not of a method for organoleptic assessment.

How the assessment is performed and how its result is interpreted therefore remains at the discretion of the laboratory. As a consequence, an organoleptic verdict may by itself determine the non-compliance of a batch, even when all other determinations raise no objections.

The psychophysical literature points to two independent sources of variability in organoleptic assessment. The first is the interindividual, partly heritable variability in the perception of basic tastes: the recognition threshold for sourness is about 53% heritable [8], and polymorphisms of the *GNAT3* gene explain about 13% of the variance in the perception of sucrose sweetness [9]. The same sample therefore evokes sensations of genuinely different intensity in different people, and perceptual experiences cannot be compared directly across individuals [10]. The second source is the non-comparability of descriptive scales: verbal labels (e.g., "very strong", "typical") denote different absolute intensities for different people, which makes category scales an inappropriate tool for comparisons between individuals and groups [11].

The aim of the study was to test empirically whether the assessment of honey taste and odor typicality is reproducible enough to serve as a stand-alone criterion of commercial quality. To this end, the results of a blind consumer panel were compared with the organoleptic assessments of the same samples carried out independently by laboratories. The analysis focused on the ability of the blind consumer panel to discriminate between samples in terms of taste typicality, on the agreement of the assessments between laboratories, and on the extent to which consumers and laboratories identified the same samples as not typical.

2. Materials and Methods

2.1. Honey Samples

Seven honey batches, labeled P1–P7, were examined. For all batches, the results of physicochemical and microscopic analyses performed before the present study were available. All samples met the requirements for the commercial quality of honey [3]. The detailed results of these analyses are not the subject of this paper and are available from the authors. The seven batches were then assessed organoleptically by two independent methods: (i) in a blind consumer panel ($n = 138$ respondents) (Section 2.2) and (ii) in an organoleptic assessment commissioned in parallel from four laboratories (Section 2.3). Three research questions were posed: (1) whether the blind consumer panel discriminates between samples in terms of taste typicality; (2) whether independent laboratories agree with one another on the typicality of the same samples; (3) whether the panel and the laboratories identify the same samples as not typical, and how strongly.

2.2. Blind Consumer Panel

Panelists were recruited at random from persons meeting the inclusion criteria: no allergy or hypersensitivity to bee products and no temporary or permanent disorders of smell and taste. The samples were coded and served in random order, without information on the variety, producer or origin. Standardized assessment conditions were ensured (serving temperature, lighting, palate cleansing between samples). Each panelist assessed all seven samples.

The assessment sheet, consisting of single-choice questions, covered the panelist's data (age group: <25, 25–45, 46–65, >65 years; gender; frequency of honey consumption: daily or several times a week, once a month, occasionally) and, for each sample: (i) the rating, i.e., overall acceptance of the taste on a scale from 1 (unacceptable) to 5 (I like it very much); (ii) sweetness, (iii) acidity and (iv) aroma (intensity), each on the scale too weak / just right / too strong; (v) typicality, i.e., the judgment whether the taste is typical of natural honey (yes / no). Question (v) corresponds to the "typicality" criterion used in the laboratory assessment and is the main comparison variable. The sheet also included a field for free comments.

2.3. Organoleptic Assessment in the Laboratories

After the panel had been completed, the same seven batches were submitted in August 2026 (10–17 August 2026) in parallel to four laboratories: Quality Services International GmbH (Bremen, Germany), Eurofins Food Integrity Control Services GmbH (Ritterhude, Germany), Intertek Food Services GmbH (Bremen, Germany) and the National Institute of Horticultural Research, Apiculture Department in Puławy (Poland), referred to in this paper as Lab1, Lab2, Lab3 and Lab4, respectively; the numbering is arbitrary. The laboratories received the samples as standard commercial orders, without coding with respect to the commercial identity of the samples and without knowledge of the other orders. The organoleptic assessment was performed according to the laboratories' own methods: ASU L 00.90-6 (modified), ICS SOP 520-02, DIN 10964 (accredited method), and PN-88/A-77626 [12] combined with the method laid down in the regulation on analytical methods for honey [6] (non-accredited method). Only the results of the organoleptic assessment (odor and taste) were used in this paper.

Lab1, Lab2 and Lab3 described odor and taste with descriptive methods, supplementing the description with a typicality qualifier. None of the reports defined this qualifier or indicated a reference standard, and its wording points to two frames of reference. The term "typical for honey" (*honey-specific*, *honigtypisch*) relates the sample to the characteristics of natural honey; the laboratories linked the opposite rating to non-compliance with Directive

2001/110/EC or to a recommendation to test for adulteration. The term "typical for the source" (*source-specific*, *trachttypisch*) relates the sample to the botanical source established by pollen analysis and served as the basis for the varietal name; in this respect, Lab2 referred to the German guidelines for honey [13]. Reference to the botanical source prevailed in positive ratings, whereas negative ratings referred exclusively to honey in general. The descriptions were coded by rules on a three-level scale, without distinguishing between the frames of reference: T (typical), L (slightly or weakly typical), N (not typical for honey). The result of a laboratory for a sample was the lower of the odor and taste scores (if no qualifier was given for one of the attributes, the score of the other was used), and an N rating was treated as an organoleptic objection. Lab4 compared the characteristics of the samples with the requirements for Polish varietal honeys and did not use a typicality qualifier. Its final verdict, "ambiguous characteristics", means that the sample cannot be classified as not typical for honey, and it was therefore coded as L. All four laboratories were included on equal terms in all measures. The coding was performed automatically on the basis of the verbatim descriptions in the reports (Appendix A).

2.4. Statistical Analysis

The analyses were performed in Python (`pandas`, `NumPy`, `SciPy`). The significance level was set at $\alpha = 0.05$ and all tests were two-sided. Given the ordinal nature of the scales (1–3 and 1–5) and the binary nature of taste typicality, nonparametric methods and methods for categorical data were used. Each respondent rated all seven samples.

Blind consumer panel. For each sample, the proportion of "typical taste" ratings was calculated with the 95% Wilson interval [14], and a binomial test was used to check whether it differed from 0.5 (no predominance of either answer). The ability of the panel to discriminate between samples was assessed with Cochran's Q test for typicality [15] and with the Friedman test for overall acceptance, both for all seven samples and after excluding the sample with extremely high typicality.

Laboratories. Agreement among the four laboratories was expressed with Fleiss' κ [16] (interpreted according to [17]) on the binary scale (objection / no objection) and on the three-level scale (N / L / T).

Panel versus laboratories. A single reference was used in all comparisons: the rating N ("not typical for honey"), the only laboratory category that corresponds to the panel question on whether the taste is typical of natural honey, and the majority verdict of the laboratories (rating N in more than half of the laboratories). The comparison was carried out at three levels. (i) The panel as a whole was treated as another rater: a sample was classified as typical when the lower limit of the 95% Wilson interval exceeded 0.5, as not typical when the upper limit was below 0.5, and as inconclusive otherwise. For the samples flagged by the laboratories, the number of respondents was calculated at which the observed proportion of "typical" ratings would give an interval excluding 0.5. (ii) Agreement within pairs of raters (laboratory–laboratory, consumer–consumer, consumer–laboratory) was expressed as the proportion of pairs agreeing on the same sample, averaged over samples, and as the coefficient $\kappa = (p_o - p_e)/(1 - p_e)$, in which p_e follows from the pooled proportions of "typical" ratings in the compared groups; for pairs of laboratories this measure is equivalent to Fleiss' κ. (iii) The ability of a single panelist rating to reproduce the laboratory verdict was assessed with the area under the ROC curve (AUC) [18] with a 95% bootstrap interval (2000 resamples of respondents with replacement, which accounts for the seven ratings given by the same person), and with the error rates of the rule "the sample is not typical when the panelist rated its taste as not typical". In addition, the agreement of the sample rankings was assessed with the Spearman rank correlation between typicality in the panel and the number of laboratories without objections ($n = 7$ samples), and the flagged samples were

compared with the remaining ones at the respondent level: for each person, the difference in mean typicality (and in overall acceptance) between the two groups of samples was calculated and tested with the Wilcoxon signed-rank test [19]; the effect size was expressed as Cohen's d for paired data [20].

3. Results

3.1. Panel Characteristics

A total of **138** respondents took part in the study and gave 966 ratings in total (7 per person). According to the dates in the panel sheet, the samples were evaluated between May and July 2026. The panel comprised 71 women (51.4%) and 64 men (46.4%); gender was not reported by 3 persons. The largest age group consisted of persons under 25 years ($n = 51$; 37%), followed by persons aged 25–45 years ($n = 37$; 26.8%) and 46–65 years ($n = 36$; 26.1%); 13 persons (9.4%) were older than 65 years, and 1 person did not report age. Honey was consumed daily or several times a week by 44 persons (31.9%), once a month by 35 (25.4%), and occasionally by 55 (39.9%); 4 persons did not answer this question.

3.2. Organoleptic Profile of the Samples in the Blind Consumer Panel

The organoleptic profile of the seven samples is presented in Table 1. Mean overall acceptance was 3.261 (SD 1.241), and the proportion of "taste typical of natural honey" ratings was 0.563. Sample P3 stood out (typicality 0.905); the typicality of the other six samples fell within a narrow range, from 0.438 (P6) to 0.558 (P1).

Table 1. Organoleptic profile of the samples in the blind consumer panel. *Rating*: mean overall acceptance (1–5); *Typicality*: proportion of "taste typical of natural honey" ratings; *Sweetness*, *Acidity*, *Aroma*: mean on the 1–3 scale (1 = too weak, 2 = just right, 3 = too strong); n: number of answers to the typicality question.

Sample	n	Rating	Typicality	Sweetness	Acidity	Aroma
P1	138	3.37	0.558	2.07	1.77	1.69
P2	137	3.04	0.496	2.06	1.77	1.81
P3	137	3.92	0.905	2.20	1.89	2.06
P4	137	3.15	0.540	2.12	1.82	1.82
P5	136	3.16	0.515	2.33	1.89	2.08
P6	137	3.07	0.438	2.26	1.87	2.04
P7	137	3.11	0.489	2.23	1.81	1.85

The panel discriminated significantly between the samples both in typicality ($n = 136$ respondents with complete answers; $Q = 94.00$, $df = 6$, $p < 0.001$) and in overall acceptance ($n = 129$; Friedman $\chi^2 = 55.35$, $p < 0.001$). After excluding P3, the differences among the remaining six samples were not significant either for typicality ($Q = 6.35$, $df = 5$, $p = 0.274$) or for overall acceptance ($\chi^2 = 7.95$, $p = 0.159$). The proportion of "typical" ratings differed significantly from 0.5 for only 1 of the 7 samples (P3) (Table 2).

3.3. Organoleptic Assessment in the Laboratories

The coded verdicts of the four laboratories are listed in Table 2, and the full verbatim descriptions in Table A1 (Appendix A). Lab1, Lab2 and Lab3 rated the odor or taste of samples P6 and P7 as not typical for honey, and samples P1 and P2 were flagged by one of them. No laboratory raised objections to samples P3, P4 and P5. For all samples, Lab4 issued the verdict "ambiguous characteristics" (code L), and in its descriptions it reported an odor characteristic of signs of fermentation in P1, P5 and P6, an unnatural odor in P2 and a candy-like odor in P7. Full agreement of the four laboratories on the binary scale was

reached for 3 of the 7 samples ($\kappa = 0.30$, fair agreement), and on the three-level scale for 1 of the 7 samples ($\kappa = 0.13$, slight agreement). If the Lab4 descriptions indicating fermentation or an unnatural or candy-like odor were coded as N, κ on the binary scale would be 0.47, and the majority verdict of the laboratories would remain unchanged. Coding only the taste descriptions yielded the same majority classification of the samples as coding the lower of the odor and taste scores.

3.4. Comparison of the Panel with the Laboratory Results

Panel as a rater. Taken as a whole, the panel classified only sample P3 as typical and did not flag any sample (Table 2). For the remaining six samples, the 95% confidence interval included 0.5 and the panel verdict remained inconclusive. The laboratories flagged samples P6 and P7 by majority (3 of 4 N ratings each). At the observed proportions of "typical" ratings, the confidence interval would exclude 0.5 only with 250 respondents for P6 and with 8012 for P7.

Table 2. Blind consumer panel versus the organoleptic assessment of Lab1, Lab2, Lab3 and Lab4. *n*: number of answers in the panel; *Typicality*: proportion of "typical taste" ratings with the 95% Wilson interval; *p*: binomial test against 0.5; *Verdict*: verdict of the panel as a whole (T: interval above 0.5, N: interval below 0.5, inc.: inconclusive). Laboratory codes: T: typical; L: slightly or weakly typical (for Lab4: "ambiguous characteristics"); N: not typical for honey (lower of the odor and taste scores). *Obj.*: number of laboratories with an N rating.

	Blind Consumer Panel					Laboratories				
Sample	***n***	**Typicality**	**95% CI**	***p***	**Verdict**	**Lab1**	**Lab2**	**Lab3**	**Lab4**	**Obj.**
P1	138	0.558	0.47–0.64	0.201	inc.	N	L	T	L	1/4
P2	137	0.496	0.41–0.58	1.000	inc.	N	L	L	L	1/4
P3	137	0.905	0.84–0.94	<0.001	T	T	L	T	L	0/4
P4	137	0.540	0.46–0.62	0.393	inc.	L	L	L	L	0/4
P5	136	0.515	0.43–0.60	0.797	inc.	L	L	T	L	0/4
P6	137	0.438	0.36–0.52	0.171	inc.	N	N	N	L	3/4
P7	137	0.489	0.41–0.57	0.864	inc.	N	N	N	L	3/4

Agreement within pairs of raters. On average, pairs of laboratories agreed on whether a sample was not typical for honey in 71.4% of cases ($\kappa = 0.30$), pairs of consumers in 54.6% ($\kappa = 0.08$), and consumer–laboratory pairs in 57.5% ($\kappa = 0.10$) (Table 3). Full agreement across all seven samples was reached only by the pair Lab2–Lab3. The rating of a single consumer agreed with that of a laboratory only slightly more often than would be expected by chance, and to a similar degree as with the rating of another consumer.

Table 3. Agreement within pairs of raters in the assessment of the same samples on the scale not typical for honey / other ratings, averaged over the seven samples. κ: agreement corrected for chance agreement.

Pair of Raters	**Agreeing Pairs (%)**	κ
laboratory–laboratory	71.4	0.30
consumer–consumer	54.6	0.08
consumer–laboratory	57.5	0.10

Single rating. The rating of a single panelist reproduced the majority verdict of the laboratories poorly (AUC = 0.570, 95% CI: 0.536–0.603, against 0.5 for guessing and the commonly accepted minimum of 0.7 for acceptable discrimination [21]). The rule "the sample is not typical when the panelist rated its taste as not typical" would classify as

not typical 39.7% of the ratings of samples to which the majority of laboratories raised no objection, and would classify as typical 46.4% of the ratings of samples flagged by the majority.

Ranking and contrast. Typicality in the blind consumer panel was correlated with the number of laboratories without objections ($\rho = 0.76$, $p = 0.049$; for overall acceptance $\rho = 0.60$, $p = 0.150$) (Figure 1). After excluding sample P3, the correlation was not significant ($\rho = 0.72$, $p = 0.109$). With the mean T/L/N code instead of the number of objections, the correlation was $\rho = 0.87$ ($p = 0.010$).

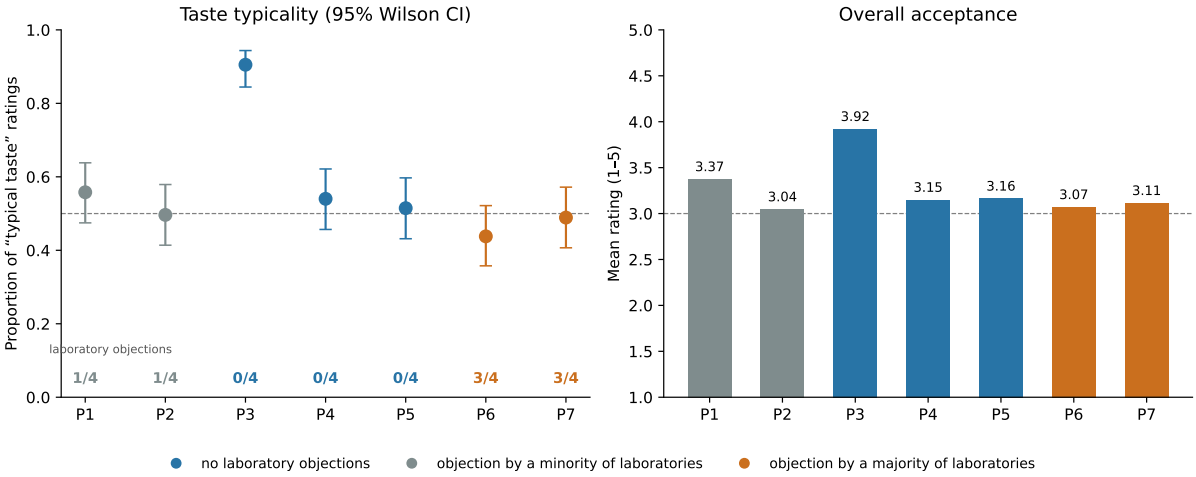


Figure 1. Taste typicality (points, 95% Wilson interval) and overall acceptance (bars) of the seven samples in the blind consumer panel. Color indicates the laboratory verdict: blue, no objections; grey, objection by a minority of laboratories; orange, objection by a majority. Numbers at the bottom of the left panel: number of laboratories (out of four) that rated the odor or taste as not typical. Dashed lines: 0.5 and 3.

The samples flagged by the majority of laboratories (P6 and P7) obtained a typicality of 0.438 and 0.489 in the panel, lower than the mean but indistinguishable from 0.5 ($p = 0.171$ and $p = 0.864$). In the paired analysis, the typicality of the remaining samples (P1, P2, P3, P4 and P5: 0.603) was higher than that of the flagged samples (0.464); the difference was $\Delta = 0.140$ (95% CI: 0.071–0.208; $p < 0.001$; $d = 0.34$), and for overall acceptance $\Delta = 0.24$ points ($p = 0.011$) (Figure 2). After excluding P3, the difference in typicality fell to $\Delta = 0.064$ and was no longer significant ($p = 0.109$).

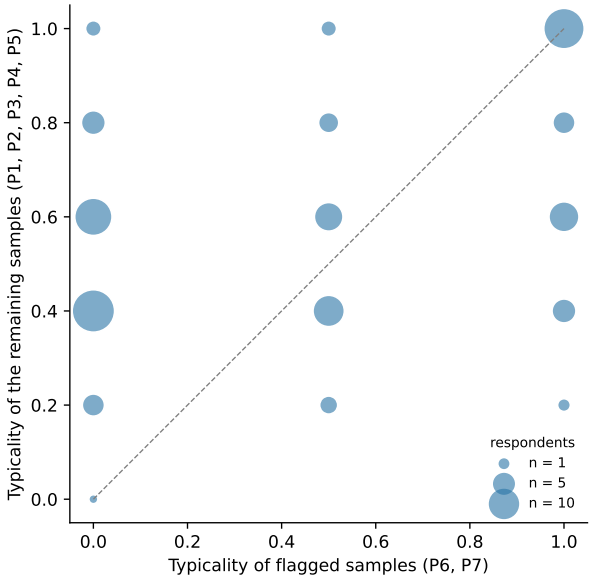


Figure 2. Mean taste typicality of the samples to which the majority of laboratories raised no objection (y-axis) and of the samples flagged by the majority of laboratories (x-axis) for each respondent; point size: number of respondents. Dashed line: equality of both ratings ($y = x$).

The panel did, however, register a feature consistent with some of the laboratory descriptions: sweetness was rated as "too strong" most often for P5 (41.3%), P6 (40.9%) and P7 (38.7%), and least often for P1 (22.1%), P3 (22.6%) and P2 (24.1%). Lab4 described the taste of P7 as "very sweet", which agrees with the panel rating, but also the taste of P2, which the panel rarely rated as too sweet.

The blind consumer panel and the laboratories ranked the samples similarly in terms of typicality, but they did not agree on which samples should be considered not typical. The typicality ranking in the blind consumer panel matches the laboratory ranking: sample P3, to which the laboratories raised no objections and which received the highest laboratory codes, was rated by far the highest by consumers, whereas samples P6 and P7, flagged by three of the four laboratories, were placed in the lower part of the ranking. Taste assessment is therefore not devoid of content, since it reflects real differences between honeys.

This agreement, however, rested on a single sample (P3), the only one classified as typical by the blind consumer panel. After its exclusion, the panel did not discriminate between the remaining six samples either in typicality or in overall acceptance, and the typicality of P6 and P7 did not differ significantly from 0.5: roughly half of the consumers considered the taste of the samples flagged by most laboratories to be typical. The difference between the flagged samples and the others was significant only because of P3 and disappeared after its exclusion. The comparison on the same scale shows this directly. The panel as a whole did not flag any sample, and at the observed typicality a decision for P6 would require about 250 respondents. A single consumer agreed with a laboratory in 57.5% of pairs ($\kappa = 0.10$), i.e., to a similar degree as with another consumer ($\kappa = 0.08$), whereas two laboratories agreed in 71.4% of pairs ($\kappa = 0.30$). Typicality assessment thus recognizes clearly typical honey but does not draw a sharp line between "typical" and "not typical" honey, which is exactly the kind of line that a criterion of compliance with commercial quality requirements needs.

4. Discussion

The consequences of low-quality organoleptic assessments have been described most thoroughly for wine. In studies of wine judges, only about 10% of them were able to

replicate their own rating of the same blind wine sample within a single category [22]. Moreover, winning a high award at one competition did not predict the result at another: the two results were statistically independent [23]. Organoleptic assessment can be calibrated through the selection and training of assessors, monitoring of panel performance indicators (repeatability, reproducibility, agreement), and the use of reference standards and externally anchored scales [24,25]. For honey, such a methodology was developed within the International Honey Commission [5], and since 1999 Italy has maintained a national register of experts in the organoleptic analysis of honey, entry into which requires completed training and a passed examination [26]. In France, the methodology for the organoleptic assessment of unifloral honeys was developed by the team of M. Gonnet (INRA, now INRAE), whose work is the origin of modern organoleptic analysis of honey [5]. In Germany, the principles of the organoleptic assessment of honey were drawn up by the German Agricultural Society (DLG): its stated aims include the characterization of unifloral honeys together with pollen analysis and the detection of the effects of overheating and prolonged storage, and the assessment scheme lists, among others, the odor and taste defects "not typical for honey", "foreign odor" and "fermenting" [27]. In both countries, organoleptic assessment complements laboratory testing rather than replacing it. In routine honey assessment, however, these indicators are usually not published in the reports, and laboratories do not share a common reference material. The comparability of organoleptic verdicts between laboratories thus remains an assumption rather than a demonstrated property of the measurement.

The same pattern is evident between laboratories. Agreement among the four laboratories was fair on the binary scale and slight on the three-level scale [17], and full agreement concerned only samples to which no laboratory raised objections. Samples P1 and P2 were flagged by only one of the four laboratories, while the others described them as typical, slightly typical or of ambiguous characteristics. Lab4, which used a different method, issued the verdict "ambiguous characteristics" for all seven samples, including P6 and P7, which were flagged by the other three laboratories, and reported an odor characteristic of signs of fermentation in three samples, including P5, to which none of the other laboratories raised objections. The verdict for the same batch therefore depends not only on random variability of perception but also on the method applied and on the laboratory.

These discrepancies do not prove an error by any particular laboratory or a defect of any particular batch. They do show, however, that in commercial quality control practice the typicality of odor and taste is not measured reproducibly between laboratories, in line with the predictions of the psychometric literature [11,23,24]. The label "not typical" results from a comparison with an internal, implicit reference of the assessor, and this reference differs between persons and laboratories.

4.1. Describing an Attribute Versus Its Decision Weight

The panel registered an attribute present in some of the laboratory descriptions: the sweetness of P6 and P7 was rated as "too strong" more often than that of other samples, and these samples received the lowest typicality. This relationship was not unequivocal, however. Consumers attributed the same sweetness intensity to sample P5, to which the laboratories using a typicality qualifier raised no objections. Sample P2, in turn, described by one of the laboratories as "very sweet", was rarely rated by consumers as too sweet.

These results suggest that an organoleptic description may capture an attribute of the sample accurately and still fail to distinguish typical from not typical honey. A possible explanation is that typicality is determined not by the intensity of a single attribute but by its consistency with the profile of a given variety; this, however, requires further research. It is therefore useful to distinguish an *accurate description* from a *decisive criterion*: an attribute

may be useful for characterizing a product and at the same time unsuitable as a stand-alone basis for an administrative decision [28].

4.2. Implications for Quality Control

From the point of view of control, what matters is not whether taste assessment carries any information at all, but whether it allows a verdict to be reproduced. A single panelist rating reproduced the majority verdict of the laboratories with AUC = 0.57 (95% CI: 0.54–0.60), and a rule based on taste typicality would be wrong in nearly every second rating. The agreement of a consumer with a laboratory ($\kappa = 0.10$) did not differ from the agreement of two consumers ($\kappa = 0.08$). The laboratories themselves fully agreed on three of the seven samples on the binary scale and on one on the three-level scale. A criterion whose result for borderline samples depends on the choice of assessor or laboratory does not satisfy the condition of reproducibility and, when used on its own, the condition of proportionality.

This difference is visible in the reports themselves. Lab2 and Lab4 described a decision rule that takes measurement uncertainty into account [29, clause 7.8.6] with respect to the limit values of measurable parameters, and Lab4 explicitly named the physicochemical parameters and pollen analysis. For the organoleptic assessment, no report stated an uncertainty or a decision rule. For instrumental methods, precision is determined in inter-laboratory studies and is used to set critical differences applied when assessing compliance with honey purity criteria [30]. For the verdict "atypical taste", no analogous quantities are known.

This does not mean that organoleptic assessment is useless. It is a valuable tool for studying preferences and for describing and improving products, and also a signal for further testing, in the role foreseen for the organoleptic test in the coordinated control plan of the European Commission [31], when the result is extreme and consistent between independent assessors, as for P6 and P7. A similar relationship has been described for trained experts: in the International Honey Commission trials, the results of assessing conformity with the profile of a unifloral honey were valid for clearly good or clearly defective samples, whereas for samples with intermediate characteristics the spread of ratings was largest, and the binary "yes/no" classification gave no valid result in 23% of cases even after three panels had been combined [5]. The results indicate that for borderline samples an organoleptic verdict should not be confirmed by a further organoleptic assessment conducted without a reference. A second laboratory using its own descriptors provides another description, not a decision: one may describe the taste as "very sweet", another the odor as "candy-like", and without a defined reference set of typicality descriptors neither description establishes atypicality. The verdict "atypical taste" should therefore be verified with reliable results, namely physicochemical and microscopic determinations and, where adulteration is suspected, analytical methods of known precision [30], before it becomes the basis for a decision to withdraw a product. The role of organoleptic assessment in commercial quality control should be limited to detecting the atypical characteristics defined in the legislation, such as a foreign taste or odor and signs of fermentation [1,6], and should not extend to judging typicality, for which no reference exists.

Such a limitation corresponds to the structure of EU law. Directive 2001/110/EC formulates the organoleptic requirements as prohibitions: honey, with the exception of baker's honey, must not have any foreign taste or odor and must not show signs of fermentation [1]. An assessment of appearance, odor, taste, consistency, crystallization and visible signs of fermentation thus makes it possible to indicate a possible non-compliance and to justify referring the sample for further laboratory testing. It was in this preliminary and complementary role, and not as a stand-alone analytical method, that the organoleptic test was used in the coordinated control plan of the European Commission [31].

4.3. Limitations of the Study

The main limitation is the small number of batches. Although the panel provided 966 ratings, the sample-level comparisons (panel–laboratory correlation, agreement of verdicts) are based on seven units. Second, the panel consisted of consumers after a short briefing rather than of selected assessors; this reflects the market perspective but increases the spread of ratings. The panel question on taste typicality is also not identical to the laboratory procedures, which assess odor and taste jointly according to their own references. Third, the laboratory assessment was carried out as commercial orders rather than as a formal proficiency test: the samples were not coded, the laboratories had no common reference material, and the coding of their descriptions on the T/L/N scale, including the assignment of code L to the verdict "ambiguous characteristics", is a simplification made by the authors. Fourth, the panel (May–July 2026) and the laboratory assessment (August 2026) were separated by one to three months, during which storage-dependent characteristics (crystallization, fermentation) may have changed. Finally, only seven batches were examined, so the results should not be generalized to all honeys.

5. Conclusions

The assessment of honey taste and odor typicality conducted as in this study (by a blind consumer panel or by a single laboratory, without a common reference standard and without a reported measure of assessor agreement) is too variable to serve as a decisive parameter in the assessment of the commercial quality of a sample.

Three independent findings support this conclusion. First, the discriminative ability of the assessment is unacceptably low: a single panelist rating reproduced the laboratory verdict with AUC $= 0.57$, i.e., only slightly better than a random binary choice (AUC $= 0.5$) and clearly below the value of 0.7 regarded as the minimum for acceptable discrimination [21]. In practice, the rule "atypical taste, sample rejected" would reject 39.7% of the ratings of samples to which the majority of laboratories raised no objections, and would pass 46.4% of the ratings of samples flagged by the majority, so it would be wrong in nearly every second decision. Second, apart from one clearly typical sample, the panel did not discriminate between samples and, as a whole, flagged none of them: the typicality of the samples flagged by most laboratories (0.438 and 0.489) was indistinguishable from 0.5. Third, the assessors agreed neither among themselves nor with the laboratories: agreement among the four laboratories was fair on the binary scale ($\kappa = 0.30$) and slight on the three-level scale ($\kappa = 0.13$), and a single consumer agreed with a laboratory to a similar degree as with another consumer ($\kappa = 0.10$ and $\kappa = 0.08$).

A decision to accept a batch as compliant or to withdraw it, based solely on such a result, would depend more on who assesses the sample and where than on the properties of the sample itself. A subjective test should therefore not serve as the basis for a binary "good/bad" judgment. Organoleptic assessment retains its value as a descriptive tool and as a signal for further testing, especially when the result is extreme and consistent between independent assessors. It should not, however, decide on its own whether honey complies with the commercial quality requirements. In the light of these results, the distinction between identifying actual quality non-compliance and the subjective evaluation of the organoleptic properties of honey is of central importance. The absence of clearly defined and objective typicality criteria limits the possibility of using the individual impressions of an assessor as the basis for administrative decisions. In particular, terms such as "too sweet", "insufficiently aromatic" or "insufficiently typical" do not in themselves constitute evidence of a breach of quality requirements. Declaring a product non-compliant on the basis of such assessments, without demonstrating a link to a specific, legally defined

criterion or to a recognized or validated test method, may therefore raise serious doubts as to the objectivity and legitimacy of the decision [32, Art. 34].

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Abbreviations

The following abbreviations are used in this manuscript:

AUC	area under the receiver operating characteristic curve
CI	confidence interval
DLG	German Agricultural Society (Deutsche Landwirtschafts-Gesellschaft)
IHC	International Honey Commission
L	slightly or weakly typical (laboratory code)
N	not typical for honey (laboratory code)
ROC	receiver operating characteristic
SD	standard deviation
T	typical (laboratory code)

Appendix A. Verbatim Odor and Taste Descriptions from the Laboratories

Table A1 lists the verbatim odor and taste descriptions reported by the laboratories (August 2026, Section 2.3), which form the basis of the coding in Table 2.

Table A1. Verbatim odor and taste descriptions reported by the four laboratories for the same honey batches (August 2026). Lab1–Lab3: wording of the original test reports (Lab1: English part of the bilingual German/English report); Lab4: translated from Polish by the authors. The laboratory labels (Lab1–Lab4) follow Table 2.

Sample	Lab.	Odor	Taste
P1	Lab1	not honey-specific	not honey-specific
	Lab2	animal urine-like, slightly source-specific	sweet, bloomy, mild, slightly source-specific
	Lab3	typical for honey	sweet, fruity

Table A1 (*cont.*)

Sample	Lab.	Odor	Taste
	Lab4	weak, slightly sour, characteristic of signs of fermentation	sweet, mild
P2	Lab1	not honey-specific	not honey-specific
	Lab2	animal urine-like, slightly source-specific	sweet, bloomy, mild, slightly metallic, slightly source-specific
	Lab3	slightly source-specific	sweet, fruity, slightly source-specific
	Lab4	unnatural, slightly floral	very sweet, prickling
P3	Lab1	source-specific	source-specific
	Lab2	slightly source-specific	sweet, bloomy, herbal, slightly source-specific
	Lab3	typical for honey	sweet, bloomy, fruity, herbal
	Lab4	strong, similar to beeswax	sweet, somewhat sharp
P4	Lab1	weakly source-specific	weakly source-specific
	Lab2	slightly honey-specific	sweet, fruity, medical, slightly metallic, slightly honey-specific
	Lab3	slightly source-specific	sweet, fruity, minty, slightly source-specific
	Lab4	faint, of rapeseed flowers	sweet
P5	Lab1	weakly source-specific	weakly source-specific
	Lab2	slightly source-specific	sweet, bloomy, medical, slightly bitter, minty, slightly source-specific
	Lab3	typical for honey	sweet, fruity, minty
	Lab4	distinct, characteristic of signs of fermentation	sweet, mild
P6	Lab1	not honey-specific	not honey-specific
	Lab2	absent	sweet, no specific floral aroma, not typical for honey
	Lab3	not typical for honey	sweet, fruity, malty, not typical for honey
	Lab4	strong, characteristic of signs of fermentation	sweet, reminiscent of burnt caramel
P7	Lab1	not honey-specific	not honey-specific
	Lab2	not typical for honey	sweet, no specific floral aroma, slightly metallic, not typical for honey
	Lab3	not typical for honey	sweet, fruity, malty, not typical for honey
	Lab4	strong, candy-like	very sweet, slightly prickling

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